U.S. Treasury Yield-Curve Factor Research

Historical factor decomposition, calendar regime comparisons and transaction-cost sensitivity

RESEARCH ONLY: Fed fitted-yield observations are generally published weekly and may be revised. These results do not reconstruct the information available on historical trading dates. They are not executable returns.

Executive findings

Complete records: 9,044, 1990-01-02 to 2026-03-27. PC1: 83.9% of daily-change variance; PCs 1–3 jointly: 97.9%.

Model observations: 8,789 after warm-up; active modeled dates: 3,894; entry events: 92. All sensitivity units are hypothetical dollars per assumed $1/bp gross DV01.

Observed fitted yields and factor loadings

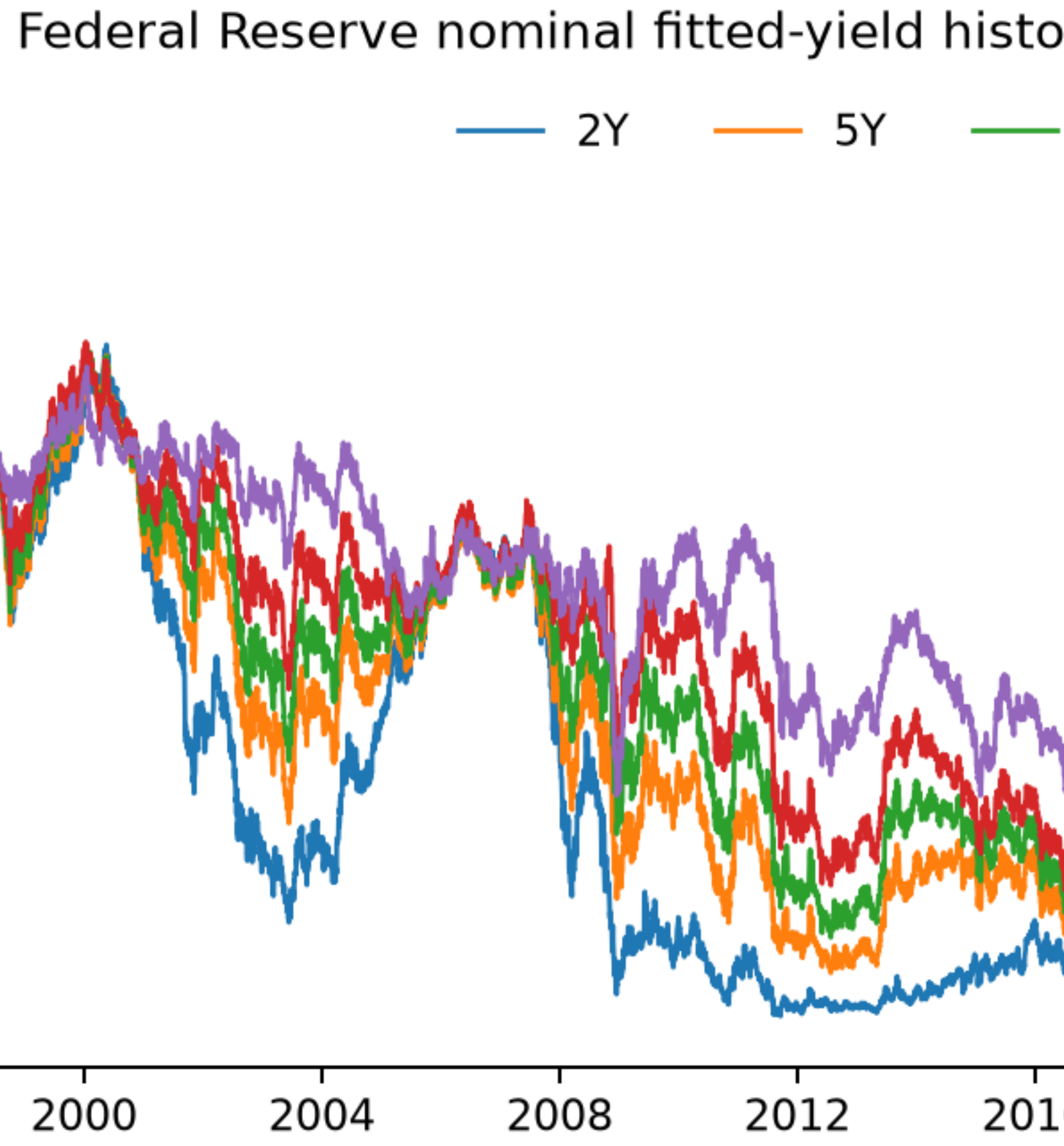


Figure 1. Model-derived nominal fitted yields, not cash-bond executable quotes.

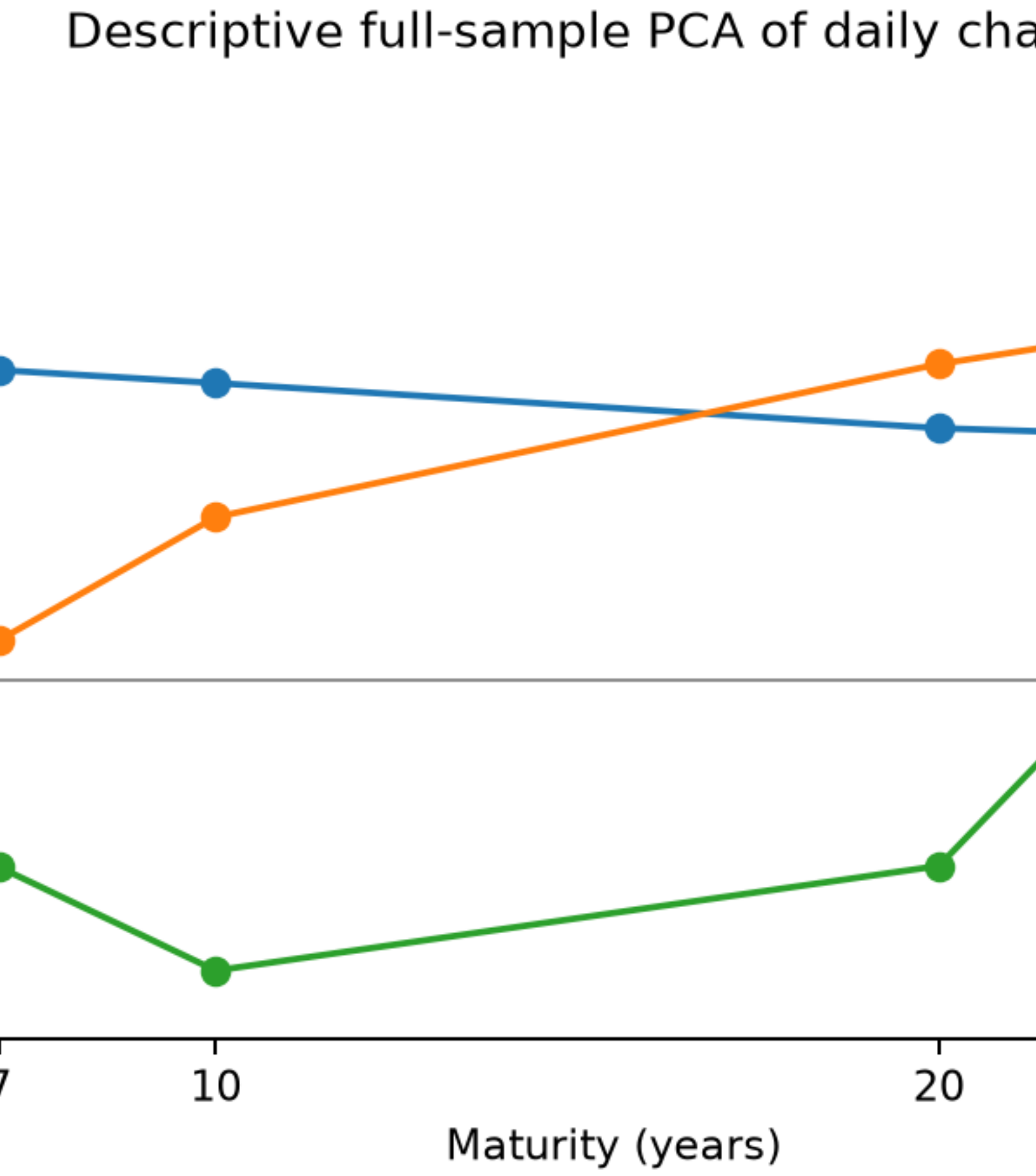


Figure 2. Descriptive PCA loadings; mathematical signs are arbitrary.

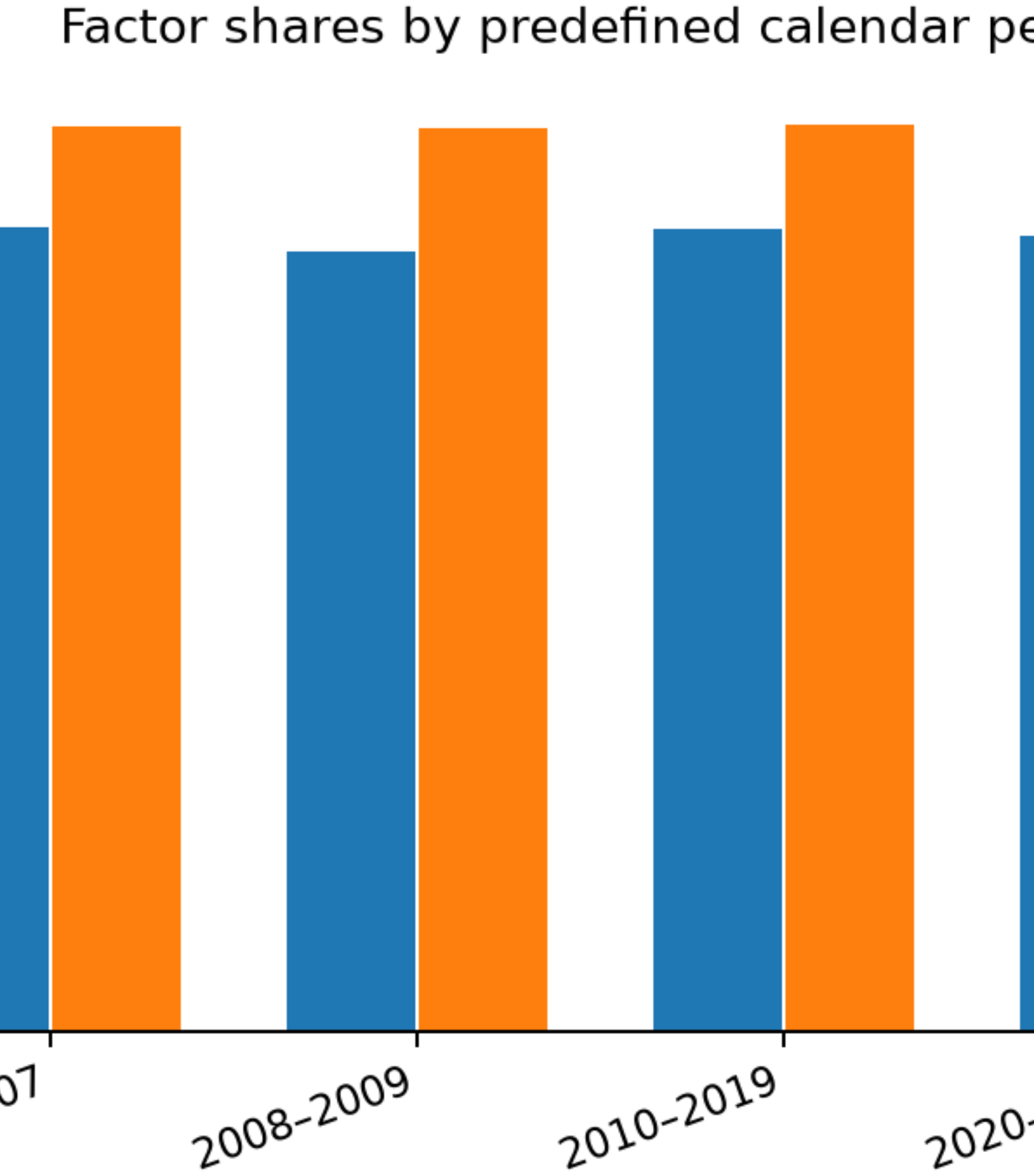

Factor shares by predefined calendar pe
07
2008–2009
2010–2019
2020-

Figure 3. Variance share by predefined historical interval.

Hypothetical sensitivity and assumed costs

A separate rolling 5Y–7Y–10Y minimum-variance fly model uses a 252-row PCA window, 126-row z-history, entry threshold 2.0 and exit threshold 0.5. Signals are shifted in observation rows; actual public-release availability is not modeled.

Assumed cost (bp)	Gross $	Cost $	Net $
0.00	5.88	0.00	5.88
0.25	5.88	47.49	-41.61
0.50	5.88	94.99	-89.10
1.00	5.88	189.97	-184.09
2.00	5.88	379.94	-374.06

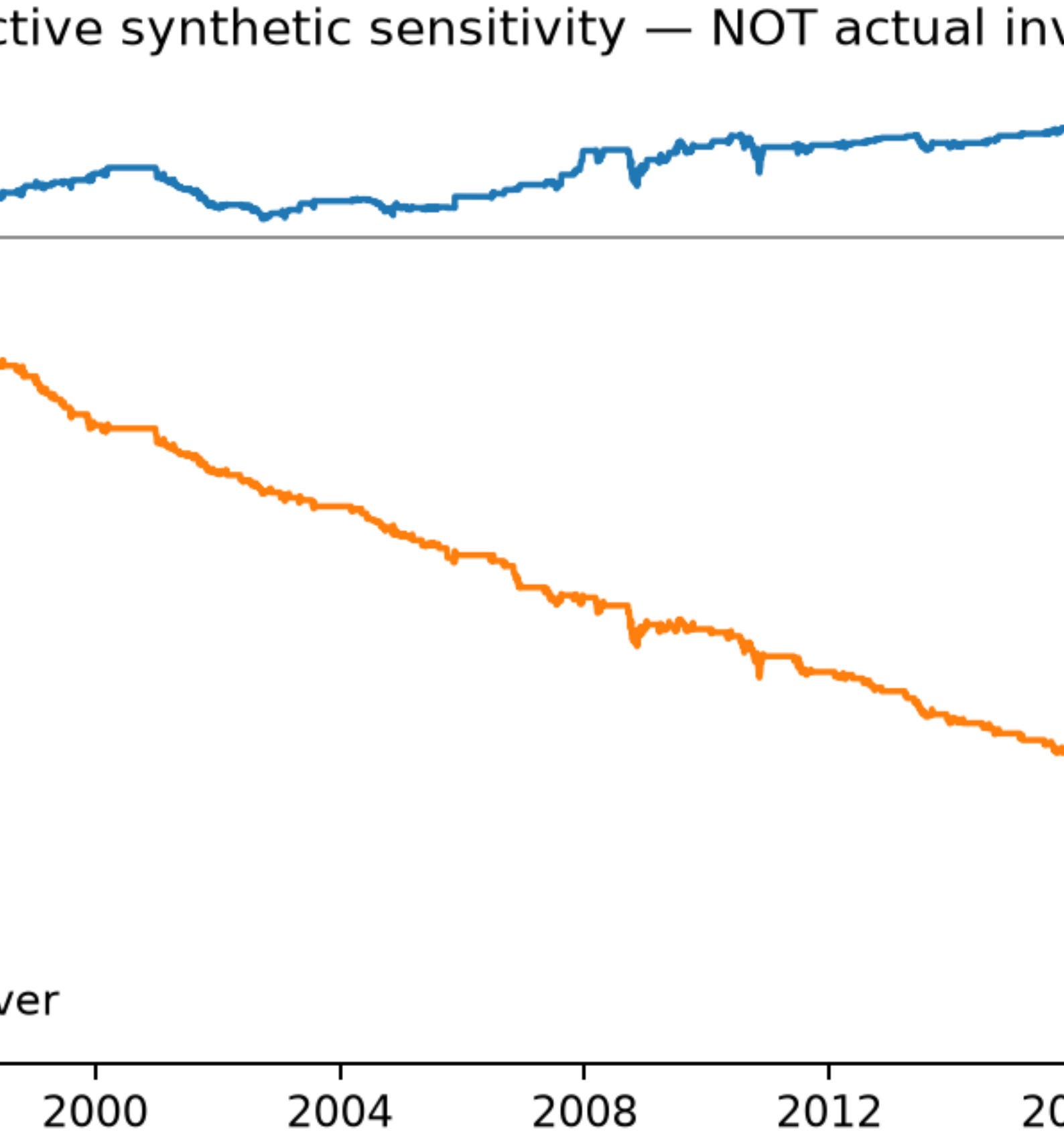


Figure 4. Retrospective synthetic cumulative sensitivity; NOT a real-bond P&L; series.

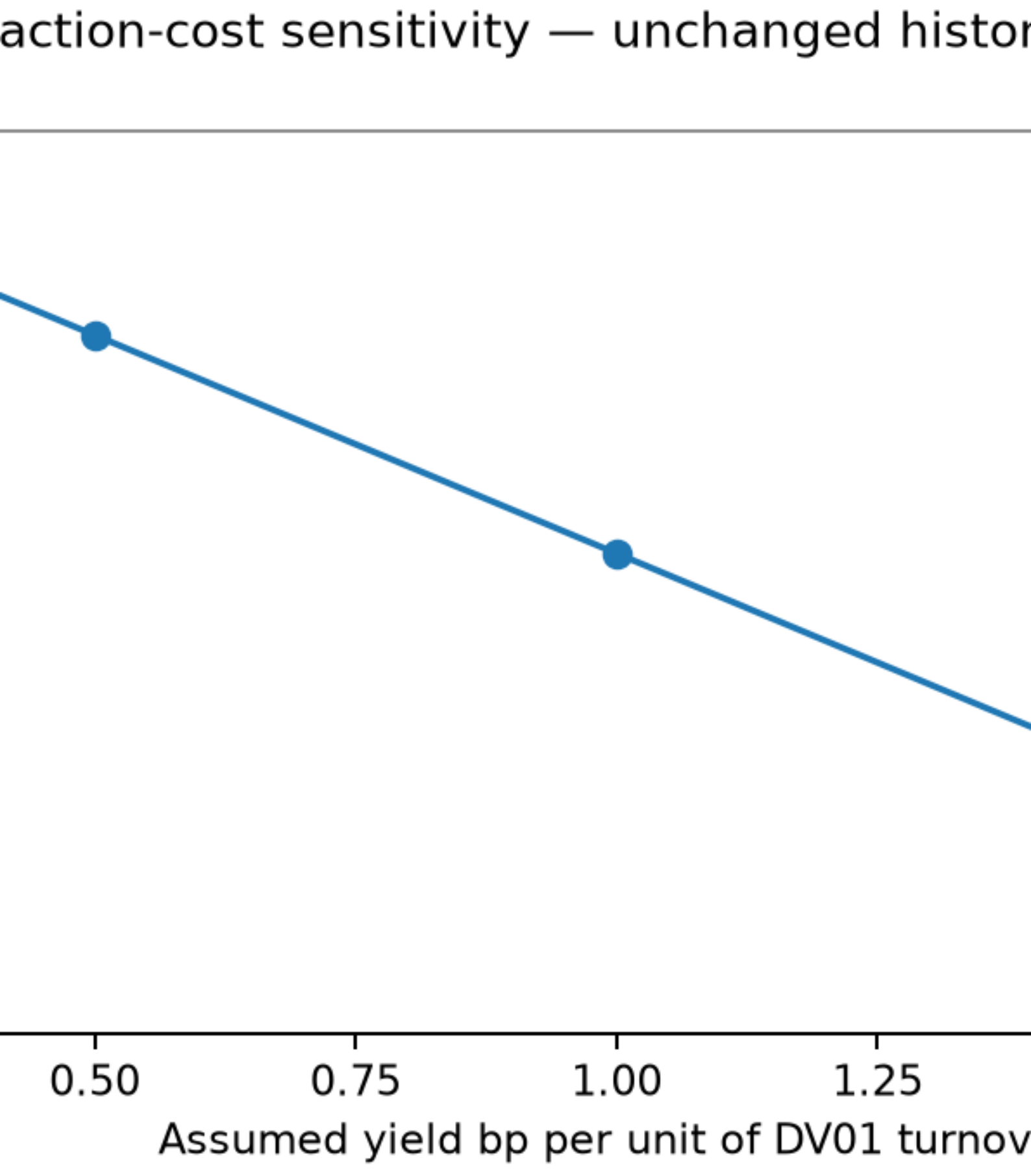


Figure 5. Net sensitivity under different hypothetical unit-turnover costs.

Limitations and sources

Data are current-vintage Fed fitted nominal yields. No historical release timestamps, live bond CUSIPs, coupon carry, financing, executable bid/ask, or investment capital are modeled. Cross-era summaries are descriptive, not out-of-sample validation.

CSV SHA-256: e43e550f1c4e360512c455096cfe350a5737129a8c51147b8803fe383a114539

Source: https://www.federalreserve.gov/data/nominal-yield-curve.htm